The RSOPT Project

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Part III

Convex Programs

Chapter 5

Linear Programs

Linear programs (`LinearProgram<N, E, I>`) are convex programs of the form

$$\min_{x \in \mathbb{R}^n} c^\top x \tag{5.1}$$

$$\text{subject to } Ax \succeq b \tag{5.2}$$

$$Ex = d \tag{5.3}$$

$$x \succeq 0 \tag{5.4}$$

where $c \in \mathbb{R}^n$, $A \in \mathbb{R}^{i \times n}$, $b \in \mathbb{R}^i$, $E \in \mathbb{R}^{e \times n}$, and $d \in \mathbb{R}^e$. As of this writing, the RSOPT project implements only the **primal-dual interior point method** (PDIPM) for solving linear programs. We acknowledge that the simplex method is a very common method for solving linear programs that converges very well in practice—however, we have selected PDIPM for its polynomial-time convergence guarantee. We begin by writing the LP (5.1) as an optimization over the relevant Lagrangian.

$$\min_{x \in \mathbb{R}^n} \max_{\lambda \in \mathbb{R}^i, \eta \in \mathbb{R}^e} c^\top x + \lambda^\top (b - Ax) + \eta^\top (d - Ex) \tag{5.5}$$

$$x \succeq 0 \tag{5.6}$$

$$\lambda \succeq 0 \tag{5.7}$$

The first-order stationarity condition associated with this problem is

$$c - A^\top \lambda - E^\top \eta = 0, \tag{5.8}$$

the equality condition for primal feasibility is

$$Ex - d = 0, \tag{5.9}$$

and the relaxed complementary slackness condition is

$$\lambda \odot (Ax - b) = \mu \mathbf{1}. \tag{5.10}$$

From these three equality conditions, we can determine a linear system for a primal update Δx, a dual update $\Delta\lambda$, and a new Lagrange multiplier vector η. We will then apply these updates to iterates x_k, λ_k to obtain iterates x_{k+1}, λ_{k+1}.

$$c - A^\top(\lambda + \Delta\lambda) - E^\top\eta = 0 \tag{5.11}$$
$$E(x + \Delta x) = d \tag{5.12}$$
$$(\lambda + \Delta\lambda) \odot (A(x + \Delta x) - b) = \mu\mathbf{1} \tag{5.13}$$

Expanding, we drop the term $\Delta\lambda \odot \Delta x$ and determine a KKT system in its canonical form.

$$\begin{bmatrix} 0 & E^\top & A^\top \\ E & 0 & 0 \\ \mathrm{diag}(\lambda)A & 0 & \mathrm{diag}(Ax - b) \end{bmatrix} \begin{bmatrix} \Delta x \\ \eta \\ \Delta\lambda \end{bmatrix} = \begin{bmatrix} c - A^\top\lambda \\ d - Ex \\ \mu\mathbf{1} - \lambda \odot (Ax - b) \end{bmatrix} \tag{5.14}$$

We write the components of the vector on the right-hand side of this relation as p, q, and r for brevity. We also define the following matrices.

$$Z := A^\top \mathrm{diag}(Ax - b)^{-1} \mathrm{diag}(\lambda) A \tag{5.15}$$
$$Y := (Z + \delta I)^{-1} \tag{5.16}$$

Here, δ is a small positive constant chosen by the user which regularizes Z and enables it to be inverted when it is rank-deficient. Solving this linear system analytically, we have the following.

$$\eta = \left(EYE^\top\right)^{-1} \left(q + EY[p - A^\top \mathrm{diag}(Ax - b)^{-1} r]\right) \tag{5.17}$$
$$\Delta x = Y\left(E^\top\eta + A^\top \mathrm{diag}(Ax - b)^{-1} r - p\right) \tag{5.18}$$
$$\Delta\lambda = \mathrm{diag}(Ax - b)^{-1}\left(r - \lambda \odot A\Delta x\right) \tag{5.19}$$

To update x_{k+1} and λ_{k+1}, we combine two techniques to ensure that iterates remain feasible. The first is a *fraction-to-the-boundary* technique that determines the maximum possible primal step size α and the maximum possible dual step size β. For each x_i and λ_i, we impose $\alpha \leq 1$, $\beta \leq 1$ as well as the constraints

$$\alpha \leq -\frac{\tau x_i}{\Delta x_i} \quad \text{where } \Delta x \leq 0, \tag{5.20}$$
$$\alpha \leq -\frac{\tau A x_i}{A\Delta x_i} \quad \text{where } A\Delta x_i \leq 0, \tag{5.21}$$
$$\beta \leq -\frac{\tau \lambda_i}{A\Delta\lambda_i} \quad \text{where } \Delta\lambda_i \leq 0. \tag{5.22}$$

Here, $0 < \tau < 1$ is a user-specified hyperparameter. The second technique selects the largest feasible α, β and performs a simple line search to ensure feasibility, scaling the step

sizes by user-selected $0 < \kappa < 1$ on each iteration until the next iterate is strictly feasible. Although this should not be necessary in theory, in practice this improves the numerical stability of the solver, as ensuring strict feasibility is critical throughout PDIPM. Subsequent iterates are computed using

$$x_{k+1} = x_k + \alpha \Delta x_k \tag{5.23}$$

$$\lambda_{k+1} = \lambda_k + \beta \Delta \lambda_k \tag{5.24}$$

and the algorithm continues until a stopping condition is reached. In Algorithm 1 we summarize PDIPM for a linear program.

Algorithm 1 Primal-Dual Interior Point Method for Linear Programming

Require: c, A, b, E, d

$x \leftarrow \mathbf{1}$
$\lambda \leftarrow \mathbf{1}$
$\eta \leftarrow \mathbf{0}$
while stopping condition not reached **do**
    $Z \leftarrow A^\top \text{diag}(Ax - b)^{-1} \text{diag}(\lambda) A$
    $Y \leftarrow (Z + \delta I)^{-1}$
    $\eta \leftarrow \left(EYE^\top\right)^{-1} \left(q + EY[p - A^\top \text{diag}(Ax - b)^{-1} r]\right)$
    $\Delta x \leftarrow Y \left(E^\top \eta + A^\top \text{diag}(Ax - b)^{-1} r - p\right)$
    $\Delta \lambda \leftarrow \text{diag}(Ax - b)^{-1} \left(r - \lambda \odot A \Delta x\right)$
    compute maximum α, β
    while next iterate infeasible **do**
        $\alpha \leftarrow \kappa \alpha$
        $\beta \leftarrow \kappa \beta$
    end while
    $x \leftarrow x + \alpha \Delta x$
    $\lambda \leftarrow \lambda + \beta \Delta \lambda$
end while
return x, λ, η

Chapter 6

Quadratic Programs

This chapter covers various forms of **quadratic programs** supported by the RSOPT project. This document defines quadratic programs as convex programs whose objective has a constant Hessian with no negative eigenvalues.

6.1 Unconstrained Quadratic Programs

Unconstrained quadratic programs (`UnconstrainedQuadraticProgram<N>`) are convex programs of the form

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^\top A x + b^\top x \tag{6.1}$$

where $A \in \mathbb{R}^{n \times n}$, $A \succeq 0$, and $b \in \mathbb{R}^n$. The relevant stationarity condition is

$$Ax^\star + b = 0 \Leftrightarrow x^\star = -A^{-1}b. \tag{6.2}$$

We solve unconstrained quadratic programs using the **conjugate gradient** (CG) method.

Definition. Two vectors p_i and p_j are *conjugate* with respect to A if $p_i^\top A p_j = 0$.

CG solves an n-dimensional system in at most n iterations by selecting conjugate search directions. The gradient of the objective at iterate x_k is

$$\nabla J_k = Ax_k + b. \tag{6.3}$$

CG selects $x_0 := 0$ and $p_0 := \nabla J_0 = b$ as the first search direction. Subsequent search directions p_k ($k > 0$) must be conjugate with all preceding search directions with respect to A. We write the recurrence as

$$p_{k+1} = Ax_{k+1} + b + \sum_{i=0}^{k} \beta_i p_i \tag{6.4}$$

and constrain β_i using the definition of conjugate vectors. For all $j = 0, 1, \cdots k$, we have

$$(Ax_{k+1} + b + \sum_{i=0}^{k} \beta_i p_i)^\top A p_j = 0. \tag{6.5}$$

Expanding gives us

$$x_{k+1}^\top A^\top A p_j + b^\top A p_j + \sum_{i=0}^{k} \beta_i p_i^\top A p_j = 0. \tag{6.6}$$

Notice that, by our "inductive hypothesis", $p_0, p_1, \cdots p_{k-1}$ are already pairwise conjugate with respect to A. We can now focus only on the case $j = k$ and take all other $\beta_i = 0$.

$$x_{k+1}^\top A^\top A p_k + b^\top A p_k + \beta_k p_k^\top A p_k = 0. \tag{6.7}$$

This gives us a closed form for β_k.

$$\beta_k = -\frac{(\nabla J_{k+1})^\top A p_k}{p_k^\top A p_k} \tag{6.8}$$

Therefore, the step direction p_k for $k > 0$ is given by

$$\boxed{p_{k+1} = \nabla J_{k+1} - \frac{(\nabla J_{k+1})^\top A p_k}{p_k^\top A p_k} p_k.} \tag{6.9}$$

After selecting the step direction, it is necessary to select the step size α_k. We select α_k according to the QP

$$\min_{\alpha_k \in \mathbb{R}} \frac{1}{2}(x_k + \alpha_k p_k)^\top A (x_k + \alpha_k p_k) + b^\top (x_k + \alpha_k p_k) \tag{6.10}$$

Expanding the objective and dropping all terms not dependent on α, we have

$$\frac{d}{d\alpha_k}\left[\alpha_k p_k^\top A x_k + \frac{1}{2}\alpha_k^2 p_k^\top A p_k + \alpha_k b^\top p_k\right] = 0. \tag{6.11}$$

We find that α_k is also determined analytically as

$$\boxed{\alpha_k = \frac{-(\nabla J_k)^\top p_k}{p_k^\top A p_k}.} \tag{6.12}$$

The complete CG algorithm is detailed in Algorithm 2. For two conjugate directions p_i and p_j with respect to A, we can interpret this algorithm in terms of the eigendecomposition of A.

$$p_i^\top A p_j = p_i^\top S \Lambda S^\top p_j \tag{6.13}$$

We can interpret the algorithm as akin to taking orthogonal steps to minimize an objective of the form $x^\top \Lambda x$, where Λ is a diagonal matrix containing the eigenvalues of A.

Algorithm 2 Conjugate Gradient

Require: A, b

$x_0 \leftarrow 0$
$p_0 \leftarrow b$
$g_0 \leftarrow b$
$\alpha_0 \leftarrow -g_0^\top p_0 / p_0^\top A p_0$
$x_1 \leftarrow x_0 + \alpha_0 p_0$
for $k = 1 \cdots n-1$ **do**
    $g_k \leftarrow A x_k + b$
    $p_k \leftarrow g_k - (g_k^\top A p_{k-1} / p_{k-1}^\top A p_{k-1}) p_{k-1}$
    $\alpha_k \leftarrow -g_k^\top p_k / p_k^\top A p_k$
    $x_{k+1} \leftarrow x_k + \alpha_k p_k$
end for
return x_n

6.2 Equality-Constrained Quadratic Programs

TODO

6.3 Inequality-Constrained Quadratic Programs

TODO